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Project Title:	The Influence of Rotational Cues on Human Tilt and Translation Responses		
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Key Personnel Changes/Previous PI:	NOTE: Dan Merfeld took over as PI from Lionel Zupan in Spring 2007; for information prior to FY2007 and for FY2007 report, please see Zupan's task with the same title. For further information, please contact the Task Book Help Desk, taskbook@nasaprs.com		
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Task Description:	Sensory systems often provide ambiguous information. For example, otolith organs measure gravito-inertial force, the sum of gravitational force and inertial force due to linear acceleration. According to Einstein's Equivalence Principle, no set of linear accelerometers alone can distinguish gravitational force (which changes with head orientation during a head tilt) from inertial force (which changes with linear acceleration of the head). Therefore, the Central Nervous System (CNS) must use other sensory cues to distinguish tilt from translation. For example, the CNS can use rotational cues provided by the semicircular canals and vision. To investigate the influence of rotational cues on the neural processing of tilt and translation in humans, we propose to develop a countermeasure to adapt and/or pre-adapt astronaut canal-otolith interactions to any gravitational environment (Earth, Mars, …) by providing correlated sensory cues between the canals and the otoliths. The countermeasure we propose to develop, often referred to as Artificial Gravity, consists of centrifugation combined with intermittent motion. We will measure behavioral (eye movements such as vestibulo-ocular reflexes) as well as psychophysical responses (orientation estimation) during tilt alone, translation alone or combined tilt-translation motion in either a 1-G or hypergravic environment. Fourier analysis and modeling tools will also be used to investigate the neural processing between rotational and gravito-inertial cues. This Task addresses the following specific aims (SA) using human subjects: SA I. Using psychophysics and eye movement measurements, investigate how humans utilize semicircular canal cues to help estimate the relative orientation of gravity (monitored through perceptual tilt responses) as well as the estimate of linear acceleration (monitored through translational VOR responses). This specific aim addresses Critical Question 9.01 since SA 1 will provide baseline data for humans to evaluate if Artificial Gravity combined with intermittent motion is a valid countermeasure (SA 2). We propose a ground study to measure human responses (perceptual tilt and eye movements) during combined canal-otolith stimulation on Earth, in a 1-G environment. Our goal is to provide a perceptual measure of tilt with simultaneous eye movement recording to comprehensively investigate how the human Central Nervous System distinguishes tilt from translation. Subjects will be tested using protocols in which they passively a) tilt, b) translate, or c) tilt & translate simultaneously. SA II. Using psychophysics and eye movement measurements, investigate how humans utilize semicircular canal cues to help estimate the relative orientation of gravity (monitored through perceptual tilt responses) as well as the estimate of linear acceleration (monitored through translational VOR responses) by measuring adaptive changes induced by exposure to a hyper-G environment. This specific aim addresses both Critical Questions (CQ) 9.01 and 9.04 by evaluating if Artificial Gravity combined with intermittent motion is a valid countermeasure (CQ 9.01) and by measuring adaptive changes induced by exposure to altered-gravity (CQ 9.04). We propose to measure human perception of orientation and eye movement responses during exposure to artificial gravity. To induce adaptation, we propose to utilize intermittent motion during centrifugation: passive roll tilt in a supine orientation will dynamically stimulate both semicircular canals and otolith organs, such that both sensory cues are correlated. (The motion envelope is in an Earth-horizontal plane.) For Specific Aim #1, we expect the human perceptual responses to be somewhat consistent with those previously measured in monkeys by Angelaki and Colleagues (1999), with the canal cue essential for accurate estimation of gravity and linear acceleration, but we also expect differences in eye responses as shown by our preliminary data with a significant horizontal eye response to roll tilt at high frequencies. The GIF resolution hypothesis states that the CNS estimates gravity and linear acceleration such that their mathematical difference matches the graviceptor measurement of the GIF. Therefore, for Specific Aim #2, based on the GIF resolution hypothesis, we expect that: a) the human responses for roll tilt about an earth-vertical axis in a supine orientation with a centripetal 1-G linear acceleration (Expt 2A) to be similar as for roll tilt about an earth-horizontal axis in an upright orientation in a 1-G environment since both dynamic canal and otolith cues are the exact same (except for a static naso-occipital 1-G gravitational acceleration in 2A): such a result would validate artificial gravity combined with intermittent motion as a countermeasure. The responses may not be exactly similar during Trial 1 but we expect the CNS to have adapted by Trial 4 back to a response similar to the response in Expt. 1 (1-G environment). b) the human responses for roll tilt about an earth-vertical axis in a supine orientation with a centripetal 1.2-G (or 1.4 G) linear acceleration (Expt 2B) to be originally (Trial 1) different from responses for roll tilt about an earth-horizontal axis in an upright orientation in a 1-G environment: for example, we expect a larger vertical VOR response (Trial 1). Then we expect the CNS to adapt to this Hyper-G environment: for example, the vertical VOR response should decrease while the horizontal VOR response increases as the CNS adapts to the hyper-G environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Since all proposed experiments are ground studies, the Earth Benefits should be straight forward. Experimental data shoud provide a better insight into how the Central Nervous System separates the graviceptor measurements of gravito-inertial force into estimates of gravity and linear acceleration, or in other words, into estimates of tilt and translation. This central process is essential to a number of everyday task such as visual stabilization or postural control and its understanding is therefore essential to the understanding of these many central processes. In addition, these ground studies should also provide insights on how the Central Nervous System adapts this separation process when humans experience a new gravitational environment. Understanding this process should not only help train and condition astronauts to microgravity or to a return to Earth-gravity after a long stay in microgravity, but it should also help us understand the rehabilitation process in vestibular patients and therefore help us design better methods of rehabilitation.
Task Progress:	NOTE: Dan Merfeld took over as PI from Lionel Zupan in Spring 2007; for Task Progress information for FY2007, please see Zupan's task with the same title. For further information, please contact the Task Book Help Desk, taskbook@nasaprs.com
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Abstracts for Journals and Proceedings	Zupan LH, Merfeld DM. "Influence of visual rotational cues on the perception of lateral translation." NASA Human Research Program Investigators' Workshop, League City, Texas, February 12-14, 2007. NASA Human Research Program Investigators' Workshop, February 2007. , Feb-2007